

01

Dashboard + source audit

ECO114 Unit 8 Part B — Supply, Demand + Perfect Competition

This pack builds the **Unit 8 Core Pack: Part B**. It focuses on demand and supply, competitive equilibrium, price-taking firms, market supply, changes in supply and demand, and long-run competitive equilibrium.

A2: 100 marks 3 hours TF 16 marks with negative marking

MCQ 44 marks with negative marking Longer questions 40 marks

Source audit

- **Used:** 2026 Econ 114 Unit 7b and 8 Perfect Competition slides.
- **Used:** Tutorial 7 English problem set, especially questions on competitive markets, profit maximisation for price-taking firms, shifts and long-run adjustment.
- **Used as context:** the agreed Unit 7–8 split from our chat.
- **Excluded:** macro Unit 8, macro policy material, Unit 7 monopoly Part A except for contrast.

What this pack does

Part B is the model foundation for Unit 8. It trains you to answer graph, calculation, TF and MCQ questions on competitive markets. It also prepares you for later Part C and Part E because surplus, taxes and price controls depend on the same demand-and-supply model.

80% standard

You must be able to solve equilibrium algebra, explain surplus/shortage adjustment,

identify shifts vs movements, and explain $P = MR = MC$ for price-taking firms without notes.

02

1. Big-picture overview: why Part B matters

Unit 8 changes the focus from a single price-setting firm to a whole market with many buyers and many sellers. The central question becomes: **how does the market price emerge, and how does it coordinate plans?**

Unit 7 price-setting firm	Unit 8 competitive market
Firm faces a downward-sloping demand curve.	Each firm faces a horizontal demand curve at the market price.
Firm chooses both quantity and price combination, constrained by demand.	Firm takes price as given and chooses output.
Profit maximisation: $MR = MC$, then price from demand.	Profit maximisation: $P = MR = MC$.
Market power can create price above marginal cost.	Competition pushes price toward marginal cost.

Core memory line

Demand shows buyers' willingness to pay; supply shows sellers' willingness to accept; competitive equilibrium is where the plans of buyers and sellers are mutually consistent.

► Why is perfect competition useful if real markets are not perfect?

2. Demand and supply foundations: WTP and WTA

The slides begin with a market for **second-hand economics textbooks**. This is useful because buyers and sellers have different personal valuations.

Demand = WTP

Demand comes from students about to take the course. Each student has a maximum amount they are willing to pay for a second-hand textbook.

- Higher WTP students buy first.
- At a high price, only the strongest buyers buy.
- At a low price, more students buy.

Supply = WTA

Supply comes from students who already completed the course. Each seller has a minimum acceptable price: their reservation price.

- Low-WTA sellers sell first.
- At a low price, only sellers with low reservation prices sell.
- At a high price, more sellers are willing to sell.

Demand curve: buyers lined up from highest WTP to lowest WTP.
Supply curve: sellers lined up from lowest WTA to highest WTA.

Key trap

WTP is not the market price. WTA is not the market price. They are reservation values that help form demand and supply curves. The market price emerges from interaction between both sides.

► If a seller’s WTA is R80 and the market price is R100, will they be willing to sell?

3. Demand and supply functions

Demand

The slides use a demand function for used textbooks:

$Q_d = 40 - 2P$

This says quantity demanded is a function of price. If price rises, quantity demanded falls.

To graph demand with price on the vertical axis, rearrange to inverse demand:

$Q_d = 40 - 2P$
 $2P = 40 - Q_d$
 $P = 20 - \frac{1}{2}Q_d$

Supply

The slides use a supply function:

$Q_s = 4P - 8$

This says quantity supplied is a function of price. If price rises, quantity supplied rises.

To graph supply with price on the vertical axis, rearrange to inverse supply:

$Q_s = 4P - 8$
 $4P = 8 + Q_s$
 $P = 2 + \frac{1}{4}Q_s$

Version	Form	Use
Ordinary demand	$Q_d = f(P)$	Plug in price to find quantity demanded.
Inverse demand	$P = f(Q_d)$	Draw the demand curve or compare WTP at a quantity.
Ordinary supply	$Q_s = f(P)$	Plug in price to find quantity supplied.

Inverse supply

 $P = f(Q_s)$

Draw the supply curve or compare WTA at a quantity.

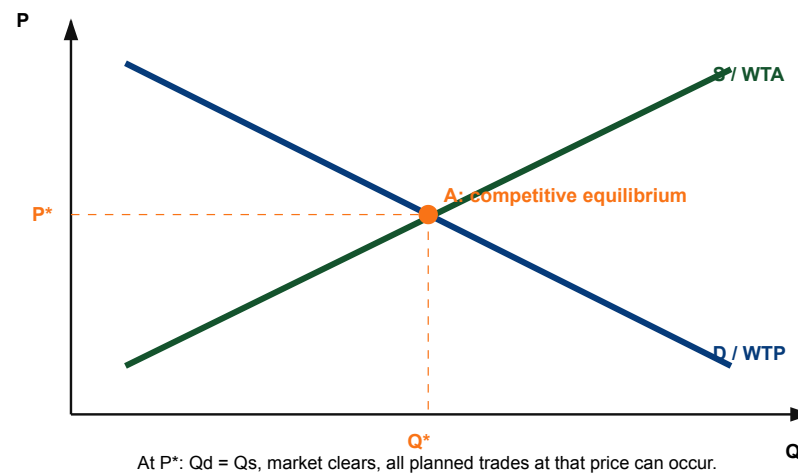
Calculation habit

Always write whether you are solving for Q_d , Q_s , P^* , or Q^* . This prevents mixing up ordinary and inverse functions.

05

4. Competitive equilibrium and market clearing

Competitive equilibrium is the price and quantity where quantity demanded equals quantity supplied.



Demand represents buyers' willingness to pay. Supply represents sellers' willingness to accept. The equilibrium price clears the market.

Method 1: set $Q_d = Q_s$

$$Q_d = Q_s$$

$$40 - 2P = 4P - 8$$

$$48 = 6P$$

$$P^* = 8$$

$$Q^* = 40 - 2(8) = 24$$

Method 2: set $WTP = WTA$

$$20 - \frac{1}{2}Q = 2 + \frac{1}{4}Q$$

$$18 = \frac{3}{4}Q$$

$$Q^* = 24$$

$$P^* = 8$$

Both methods give the same answer because the equilibrium is where marginal buyer WTP equals marginal seller WTA.

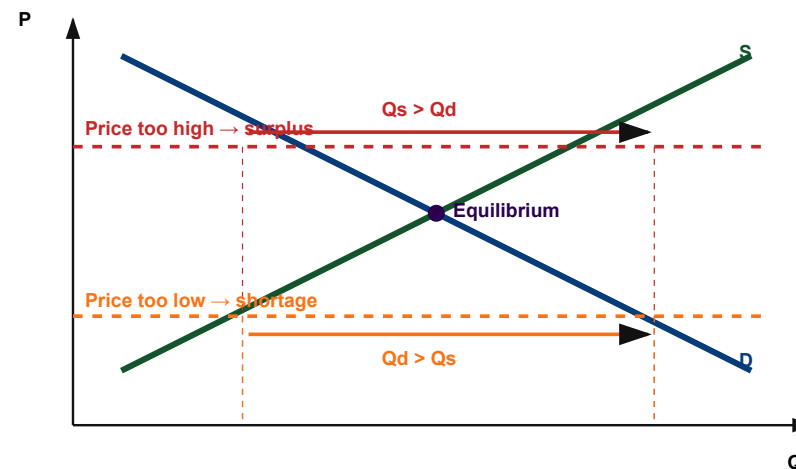
Exam wording

At the competitive equilibrium, all buyers and sellers who are willing to trade at P^* can execute their planned purchases and sales. There is no excess demand and no excess supply.

► Why is the last unit traded important?

06

5. Surplus, shortage and price adjustment



Prices adjust because surplus creates pressure to cut prices, while shortage creates pressure to raise prices.

Price too high: surplus / excess supply

Using the textbook example, if $P = \$11$:

$$Q_d = 40 - 2(11) = 18$$

$$Q_s = 4(11) - 8 = 36$$

$$\text{Surplus} = Q_s - Q_d = 18 \text{ units}$$

Only 18 units are sold because buyers limit the actual number of trades. Sellers with unsold stock have an incentive to reduce price.

Price too low: shortage / excess demand

If $P = \$5$:

$$Q_d = 40 - 2(5) = 30$$
$$Q_s = 4(5) - 8 = 12$$
$$\text{Shortage} = Q_d - Q_s = 18 \text{ units}$$

Only 12 units are sold because sellers limit the actual number of trades. Buyers unable to buy may offer more, pushing price up.

Situation	Condition	Pressure on price	Why
Surplus	$Q_s > Q_d$	Downward	Sellers compete to sell unsold stock.
Shortage	$Q_d > Q_s$	Upward	Buyers compete for scarce goods.

Precise wording

Do not say surplus means “too many buyers”. Surplus means too much supplied relative to quantity demanded at that price.

6. The model of perfect competition

Perfect competition is an idealised market structure in which both buyers and sellers are price takers.

Condition	Meaning	Why it matters
Many buyers	No individual buyer is large enough to control price.	Buyers cannot force the price down alone.
Many sellers	No individual seller is large enough to control price.	Sellers cannot raise the price without losing customers.
Homogeneous product	Products are identical/undifferentiated.	Buyers do not care which seller they buy from.
Good information	Buyers and sellers know the market price and relevant quality information.	Search/bargaining power is reduced.
Free entry and exit in the long run	Firms can enter if profit is positive and exit if loss persists.	Long-run economic profit tends toward zero.

Competitive equilibrium as a Nash equilibrium

At P^* , no single buyer or seller can improve their situation by changing behaviour alone.

- A seller charging above P^* loses buyers.
- A seller charging below P^* gains no benefit because they could sell at P^* .
- A buyer offering below P^* finds no seller.
- A buyer offering above P^* gains no benefit because they could buy at P^* .

Perfect competition memory line

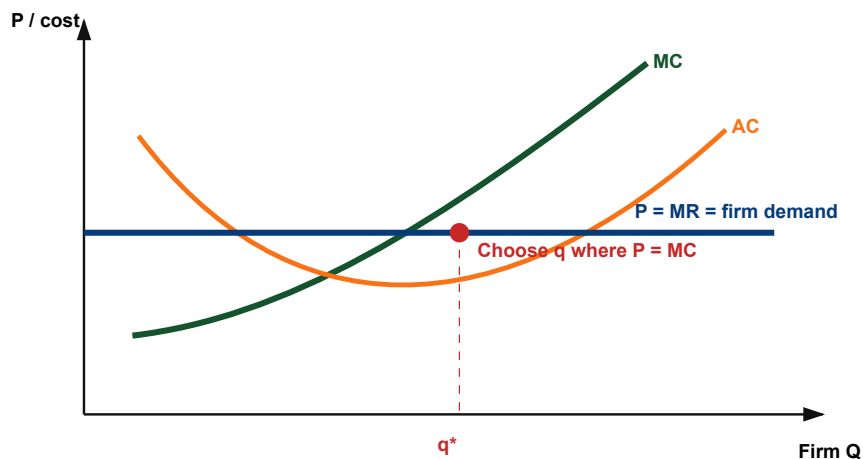
Competition eliminates bargaining power: no buyer or seller has enough power to change the market price.

► **Why does “many firms” alone not guarantee perfect competition?**

08

7. Price-taking firms: $P = MR = MC$

A price-taking firm accepts the market price as given. It can sell as much as it wants at that market price, but one small firm cannot affect the price.



A price-taking firm cannot choose price. It chooses output. Profit is maximised where $P = MR = MC$.

Key logic

For a price-taking firm: $P = MR$

Profit maximisation: $MR = MC$

Therefore: $P = MC$

If $P > MC$, producing one more unit adds more to revenue than to cost, so the firm should increase output. If $P < MC$, the last unit costs more than it adds to revenue, so the firm should reduce output.

Tutorial 7 Q2 logic

Charlie's Chimps produces 100 toys at market price R10 and $MC = R9$. Since $P > MC$, producing an extra unit adds R10 revenue but only R9 cost. The firm is not yet maximising profit; it should produce more until $P = MC$.

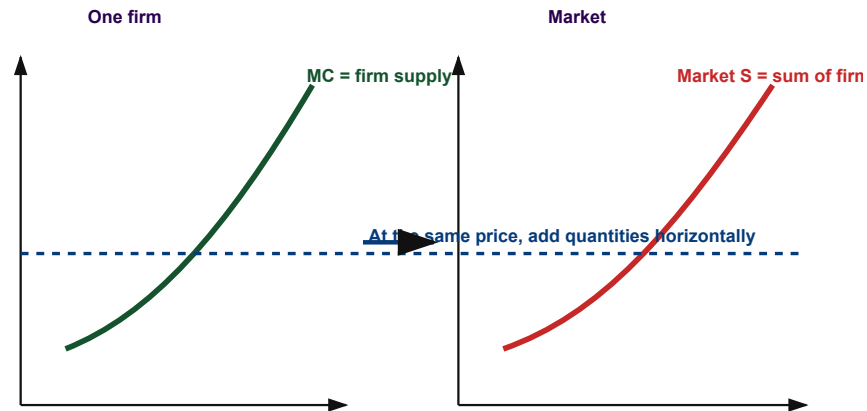
Big Unit 8 trap

A price-taking firm does not lower the price to equal MC . It cannot influence the market price. It adjusts quantity until MC equals the given market price.

► If $P = R50$ and MC at current output is $R60$, what should the firm do?

8. Firm supply and market supply

For a price-taking firm, the quantity it chooses depends on the market price. At every price, the firm chooses the quantity where $P = MC$. This means the MC curve gives the firm's supply decision, as long as the firm is producing rather than shutting down.



Market supply is the horizontal sum of individual firm supplies: at each price, add the quantities all firms choose to produce.

Market supply

The market supply curve is found by adding the quantities supplied by every firm at each price.

If 50 identical bakeries each supply 120 loaves at $P = €2.35$,
 market supply = $50 \times 120 = 6\,000$ loaves

If 50 identical bakeries each supply 66 loaves at $P = €1.52$,
 market supply = $50 \times 66 = 3\,300$ loaves

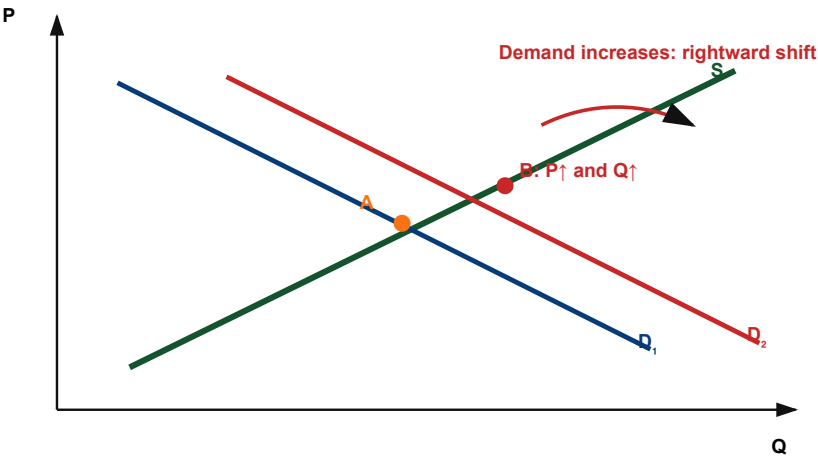
The slides describe market supply as the market's marginal cost curve: the MC of the 6,001st loaf is about the price required to get that extra loaf supplied.

Graph wording

Market supply is a horizontal summation, not a vertical addition. At the same price, add firm quantities.

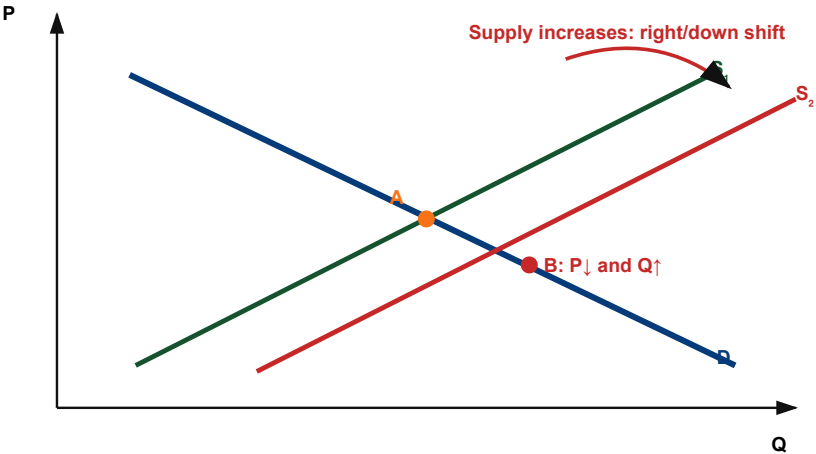
9. Changes in supply and demand

Demand shift



An increase in demand means more is demanded at every price. Equilibrium price and quantity both rise.

Supply shift



An increase in supply means more is supplied at every price. Equilibrium quantity rises and price falls.

Change	Graph effect	Equilibrium result
Demand increases	D shifts right	P rises, Q rises
Demand decreases	D shifts left	P falls, Q falls
Supply increases	S shifts right/down	P falls, Q rises
Supply decreases	S shifts left/up	P rises, Q falls

Causes from the slides

Demand increases when...	Supply increases when...
Preferences change in favour of the good.	Input prices decrease.
Price of a complement falls.	Technology reduces marginal cost.
Price of a substitute rises.	Number of firms in the market increases.

Average income or number of buyers rises.

State of nature improves, such as a good harvest.

Tutorial 7 Q3 logic

A rise in the price of spinach causes a **movement along** the supply curve, not a shift. Weather and wage changes shift supply because they affect cost/productivity at every price.

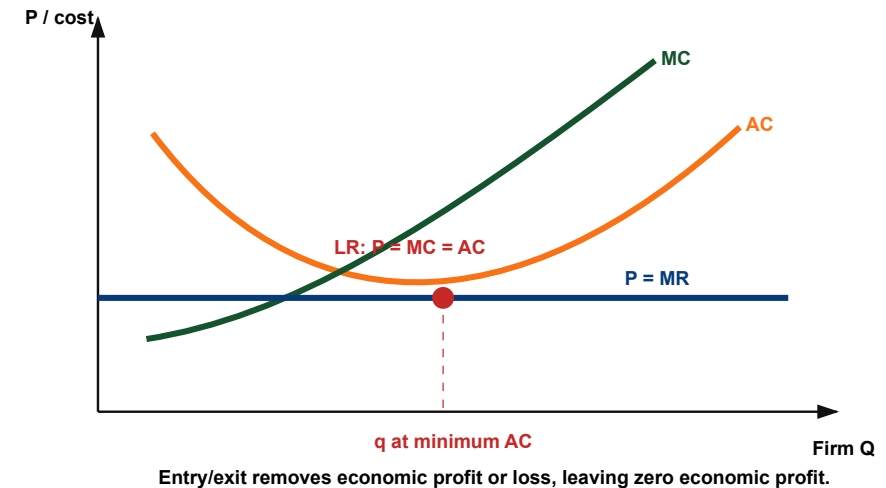
Most common MCQ trap

“Quantity supplied increases” is not the same as “supply increases”. Quantity supplied changes because price changes; supply changes when the entire curve shifts.

11

10. Short-run vs long-run competitive equilibrium

In the short run, price adjusts and firms adjust their output. Firms may make economic profit, zero economic profit, or economic losses. In the long run, entry and exit can change the number of firms.



In long-run competitive equilibrium, firms maximise profit and earn zero economic profit: $P = MR = MC = AC$ at minimum AC.

Long-run logic

1. If firms make positive economic profit, new firms enter.
2. Entry increases market supply.
3. Greater supply lowers market price.
4. Price keeps falling until economic profit is zero.
5. At long-run equilibrium, firms are still maximising profit, so $P = MC$.
6. Zero economic profit means $P = AC$.
7. Therefore: $P = MC = AC$.

Short run: $Q_d = Q_s$ and $P = MR = MC$.

Long run: $Q_d = Q_s$ and $P = MR = MC = AC$.

At long-run equilibrium, AC is minimised because $MC = AC$ at minimum AC.

Tutorial 7 Q1 logic

If firms make economic profit in a perfectly competitive market, firms enter. The market supply curve shifts out. The market price falls. Each existing firm normally produces less at the lower price, moving along its MC curve. So the statement is **True** under the standard model.

Zero economic profit ≠ no accounting profit
Economic profit already includes opportunity cost. Zero economic profit means firms earn enough to keep resources in the market, but not an above-normal rent.

11. Is perfect competition a good model?

Perfect competition is a simplified model. The slides ask whether real markets are close enough to make price-taking behaviour a useful assumption.

Two tests economists use

Test	Meaning	Why it matters
Do trades take place at the same price?	If identical goods sell at very different prices at the same time, the market may not be well approximated by one competitive price.	Competitive equilibrium predicts a single market-clearing price.
Do firms sell at a price equal to marginal cost?	If price is close to marginal cost, market power is limited.	$P \approx MC$ is a key competitive prediction.

Choccos/chocolate bars example

Even if Choccos is produced by one firm, many chocolate bars are close substitutes. Competition makes the Choccos demand curve almost flat. The firm has little market power and may set P approximately equal to MC . For many applications, modelling the wider chocolate-bar market as competitive can be reasonable.

Balanced answer
Do not answer “perfect competition is unrealistic” and stop. A good answer says whether the model assumptions are close enough to explain the outcome being studied.

12. Tutorial 7 Part B application layer

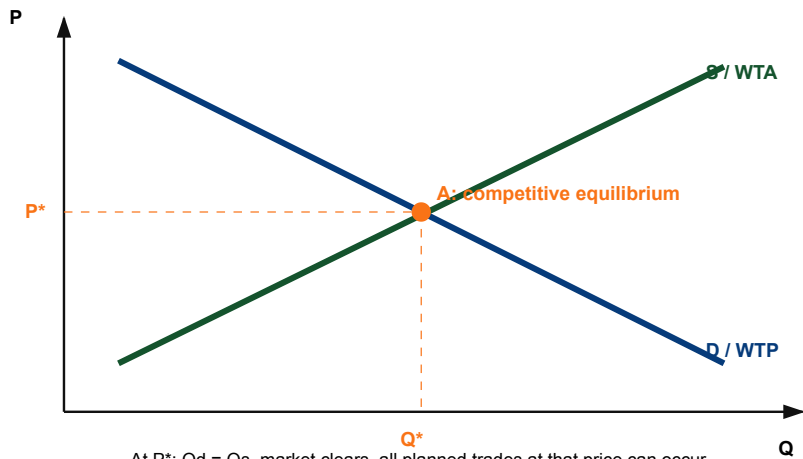
Tutorial question	Use in this Part B pack?	Model answer logic
Q1: economic profit and firm entry	Yes	True in the standard model: profit attracts entry, supply shifts out, price falls, existing firms move down their MC curves and produce less.
Q2: Charlie's Chimps, $P = R10$, $MC = R9$	Yes	Since $P > MC$, produce more. Do not lower price; the firm is a price taker.
Q3: movement along spinach supply curve	Yes	A rise in the price of spinach causes movement along supply. Weather/wages shift supply.
Q4: market-power revenue table	Mostly Unit 7 Part A	Useful contrast only; it tests downward-sloping demand/MR rather than price-taking firms.
Q5: MC/AC graph in perfect competition	Yes	If price = a, firm chooses b where $P = MC$. Profit depends on whether P is above, equal to, or below AC at b. Long run: entry/exit drives $P = MC = AC$.
Q6: monopoly inverse demand revenue effects	Unit 7 Part A/Part D	Not Part B; it tests monopoly MR gain/loss logic.

How to use tutorials

Redo Q1, Q2, Q3 and Q5 closed-book. They are exactly the kind of TF/MCQ/longer-question logic Part B can test.

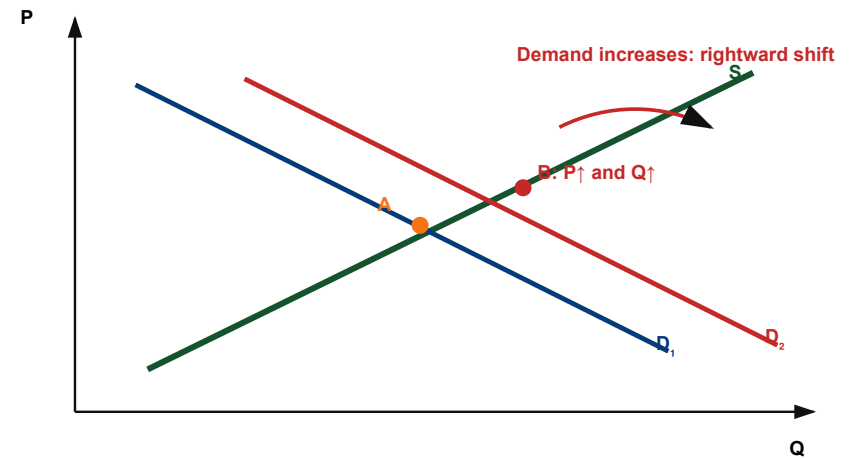
13. Graph bank: what to draw and what to say

Graph	Axes	Key label	What to write	Common mistake
Demand and supply equilibrium	P vertical, Q horizontal	D, S, P^* , Q^*	Equilibrium occurs where $Q_d = Q_s$; the market clears.	Forgetting to label P^* and Q^* .
Surplus	P vertical, Q horizontal	Price above P^* , $Q_s > Q_d$	Sellers compete to lower price.	Calling it excess demand.
Shortage	P vertical, Q horizontal	Price below P^* , $Q_d > Q_s$	Buyers compete to raise price.	Calling it excess supply.
Demand increase	P vertical, Q horizontal	D shifts right	At old price there is excess demand; new equilibrium has higher P and Q.	Saying supply increased because Q_s rose.
Supply increase	P vertical, Q horizontal	S shifts right/down	At old price there is excess supply; new equilibrium has lower P and higher Q.	Moving along supply instead of shifting it.
Price-taking firm	P/cost vertical, firm Q horizontal	Horizontal $P = MR, MC, AC$	Firm maximises profit where $P = MC$.	Letting the firm choose price.
Long-run firm equilibrium	P/cost vertical, firm Q horizontal	$P = MC = AC$	Entry/exit eliminates economic profit; price equals minimum AC.	Saying zero economic profit means the firm shuts down.

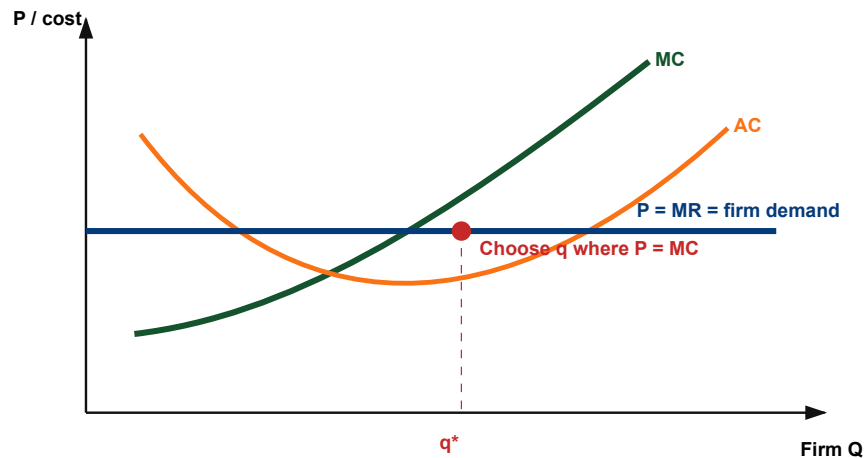


At P^* : $Q_d = Q_s$, market clears, all planned trades at that price can occur.

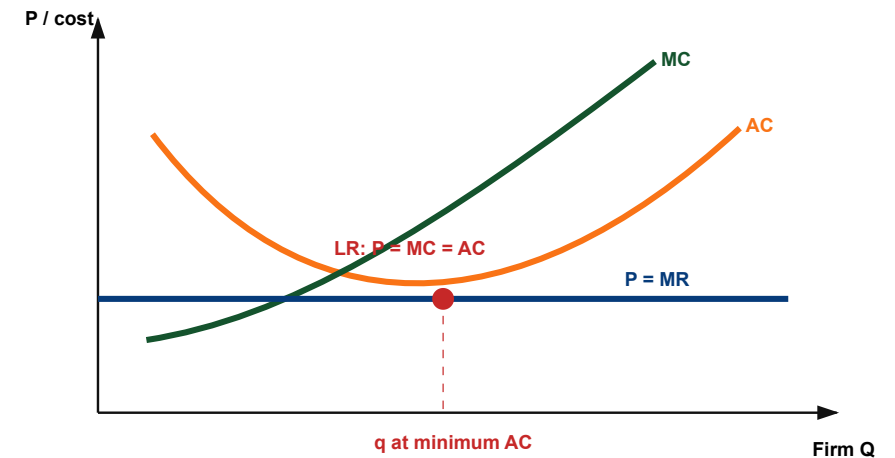
Demand represents buyers' willingness to pay. Supply represents sellers' willingness to accept. The equilibrium price clears the market.



An increase in demand means more is demanded at every price. Equilibrium price and quantity both rise.



A price-taking firm cannot choose price. It chooses output. Profit is maximised where $P = MR = MC$.



Entry/exit removes economic profit or loss, leaving zero economic profit.

In long-run competitive equilibrium, firms maximise profit and earn zero economic profit: $P = MR = MC = AC$ at minimum AC.

15

14. Formula and calculation bank

Equilibrium

$$Q_d = Q_s$$

$$\text{Example: } 40 - 2P = 4P - 8 \rightarrow P^* = 8, Q^* = 24$$

Using inverse functions

$$\text{Set } WTP = WTA$$

$$20 - \frac{1}{2}Q = 2 + \frac{1}{4}Q \rightarrow Q^* = 24, P^* = 8$$

Surplus / shortage

$$\text{Surplus} = Q_s - Q_d \text{ when } Q_s > Q_d$$

$$\text{Shortage} = Q_d - Q_s \text{ when } Q_d > Q_s$$

Price-taking firm

$$MR = P$$

$$\text{Profit-maximising firm output: } P = MC$$

Long-run competition

$$P = MR = MC = AC$$

$$\text{Economic profit} = 0$$

$$\text{AC at minimum when } MC = AC$$

Market supply

$$\text{Market } Q_s \text{ at a price} = \text{firm 1 } Q_s + \text{firm 2 } Q_s + \dots$$

$$\text{If } n \text{ identical firms each supply } q: \text{market } Q_s = n \times q$$

Algebra trap

After finding P^* , plug back into either Q_d or Q_s . If you get different quantities, your algebra is wrong because equilibrium requires $Q_d = Q_s$.

15. True/False bank for negative marking

► T/F: In a competitive equilibrium, quantity demanded equals quantity supplied.

► T/F: If price is above equilibrium, there is excess demand.

► T/F: If price is below equilibrium, sellers are normally willing to sell more than buyers want to buy.

► T/F: Willingness to pay is the maximum price a buyer would pay.

► T/F: Willingness to accept is the highest price a seller will accept.

► T/F: A rise in the price of a good causes a movement along its supply curve.

► T/F: A fall in input costs shifts the supply curve right/down.

► T/F: A price-taking firm chooses the market price where it wants to sell.

► T/F: For a price-taking firm, marginal revenue equals the market price.

► T/F: A price-taking firm maximises profit where $P = MC$.

► T/F: If P is greater than MC , the price-taking firm should decrease output.

► T/F: Market supply is found by adding firm quantities at each price.

► T/F: Market supply is found by adding prices vertically at each quantity.

► T/F: In long-run competitive equilibrium, firms earn positive economic profit.

► T/F: Zero economic profit means accounting profit must be zero.

► T/F: In long-run competitive equilibrium, $P = MC = AC$.

► T/F: At the minimum point of AC , MC equals AC .

► T/F: A demand increase shifts the demand curve right.

► T/F: A supply decrease usually raises equilibrium price and lowers equilibrium quantity.

► T/F: If demand increases, supply also increases because equilibrium quantity supplied rises.

► T/F: Perfect competition requires homogeneous products in the strict model.

► T/F: A market with differentiated products can never be approximated as competitive.

► T/F: At competitive equilibrium, no individual buyer or seller has an incentive to change behaviour alone.

► T/F: A shortage creates pressure for price to fall.

► T/F: A surplus creates pressure for price to fall.

► T/F: If firms are making losses in a competitive market, some firms may exit in the long run.

► T/F: A price-taking firm faces a horizontal demand curve at the market price.

► T/F: Perfect competition means firms have strong bargaining power over buyers.

► T/F: If the number of firms increases, market supply increases.

► T/F: Competitive equilibrium necessarily means every person who wants the good receives one.

16. MCQ bank

- ▶ MCQ 1: A market clears when:
- ▶ MCQ 2: If $Q_d = 40 - 2P$ and $Q_s = 4P - 8$, equilibrium price is:
- ▶ MCQ 3: At $P = 11$ in the same market, $Q_d = 18$ and $Q_s = 36$. This is:
- ▶ MCQ 4: At $P = 5$ in the same market, $Q_d = 30$ and $Q_s = 12$. This is:
- ▶ MCQ 5: Which event causes a movement along the supply curve for spinach?
- ▶ MCQ 6: For a price-taking firm:
- ▶ MCQ 7: A price-taking firm with $P = R10$ and $MC = R9$ should:
- ▶ MCQ 8: Market supply is the:
- ▶ MCQ 9: Perfect competition requires:
- ▶ MCQ 10: In long-run competitive equilibrium:
- ▶ MCQ 11: If firms earn positive economic profit and entry is easy:
- ▶ MCQ 12: If demand increases while supply is unchanged:
- ▶ MCQ 13: If supply increases while demand is unchanged:
- ▶ MCQ 14: WTA means:
- ▶ MCQ 15: WTP means:

- ▶ MCQ 16: At a competitive equilibrium interpreted as Nash equilibrium:
- ▶ MCQ 17: A surplus usually causes:
- ▶ MCQ 18: A shortage usually causes:
- ▶ MCQ 19: A rise in the price of wheat for bread producers will most likely:
- ▶ MCQ 20: More efficient ovens for bakeries will most likely:
- ▶ MCQ 21: In a price-taking firm graph, the horizontal line is:
- ▶ MCQ 22: If $P < AC$ in the short run:
- ▶ MCQ 23: Long-run zero economic profit occurs because:
- ▶ MCQ 24: Which is a demand shifter?
- ▶ MCQ 25: Which is a supply shifter?

17. Longer-question templates

Template 1: Explain competitive equilibrium

Opening: A competitive equilibrium is the price and quantity where quantity demanded equals quantity supplied.

1. Define demand as WTP and supply as WTA.
2. State $Q_d = Q_s$.
3. Explain that if price is above equilibrium, surplus pushes price down.
4. Explain that if price is below equilibrium, shortage pushes price up.
5. Conclude: at P^* , all buyers and sellers can execute their planned trades at the market-clearing price.

Template 2: Explain a demand increase

Opening: A demand increase means quantity demanded rises at every price, so the demand curve shifts right.

1. Draw initial D and S with equilibrium A.
2. Shift D right to D_2 .
3. At old price, $Q_d > Q_s$, creating excess demand.
4. Competition among buyers raises price.
5. New equilibrium has higher price and higher quantity.
6. Important wording: supply has not increased; quantity supplied rises along the supply curve.

Template 3: Explain price-taking firm output

Opening: A price-taking firm accepts the market price as given and chooses output to maximise profit.

1. State $MR = P$ for a price-taking firm.
2. Profit maximisation requires $MR = MC$.
3. Therefore choose q where $P = MC$.

4. If $P > MC$, produce more; if $P < MC$, produce less.
5. Profit depends on whether price is above, equal to or below AC at q .

Template 4: Explain long-run adjustment after positive profit

Opening: Positive economic profit attracts entry in a competitive market when entry is cheap and easy.

1. At the initial market price, firms earn $P > AC$.
2. New firms enter.
3. Market supply shifts right.
4. Equilibrium price falls and market quantity rises.
5. Each firm reduces output along its MC curve.
6. Entry stops when economic profit is zero: $P = MC = AC$.

Template 5: Natural disaster destroys producers

Opening: A natural disaster that destroys factories reduces the number of sellers and therefore decreases market supply.

1. Market supply shifts left/up.
2. At the old price, $Q_d > Q_s$, so there is excess demand.
3. Price rises to a new short-run equilibrium and market quantity falls.
4. Surviving firms face the higher market price and increase output until $P = MC$.
5. If surviving firms earn economic profit, entry/rebuilding may occur in the long run.
6. Long-run equilibrium depends on whether entry can restore supply; under standard free entry, profit returns to zero.

18. 50+ active recall drill

► Recall 1: Define competitive equilibrium.

► Recall 2: Define willingness to pay.

► Recall 3: Define willingness to accept.

► Recall 4: What is a reservation price?

► Recall 5: What does $Q_d = Q_s$ mean?

► Recall 6: What is market clearing?

► Recall 7: What happens when price is above equilibrium?

► Recall 8: What happens when price is below equilibrium?

► Recall 9: Why does a surplus push price down?

► Recall 10: Why does a shortage push price up?

► Recall 11: Write the demand function from the slides.

► Recall 12: Write the supply function from the slides.

► Recall 13: Convert $Q_d = 40 - 2P$ into inverse demand.

► Recall 14: Convert $Q_s = 4P - 8$ into inverse supply.

► Recall 15: Solve P^* and Q^* for the textbook example.

► Recall 16: At $P = 11$, calculate Q_d and Q_s .

► Recall 17: At $P = 5$, calculate Q_d and Q_s .

► Recall 18: List the conditions of perfect competition.

► Recall 19: Why are firms price takers?

► Recall 20: Why are buyers price takers?

► Recall 21: Why is the competitive equilibrium a Nash equilibrium?

► Recall 22: What is homogeneous product?

► Recall 23: Why does good information matter?

► Recall 24: Why does free entry/exit matter?

► Recall 25: What is the individual firm's demand curve shape?

► Recall 26: Why does $MR = P$ for price-taking firms?

► Recall 27: What output rule does a price-taking firm use?

► Recall 28: If $P > MC$, what should the firm do?

► Recall 29: If $P < MC$, what should the firm do?

► Recall 30: What is the firm supply curve in the relevant range?

► Recall 31: How do you get market supply from firm supply?

► Recall 32: What is horizontal summation?

► Recall 33: Name two causes of demand increase.

► Recall 34: Name two causes of supply increase.

► Recall 35: Does own price change shift demand?

► Recall 36: Does own price change shift supply?

► Recall 37: What happens to P and Q when demand increases?

► Recall 38: What happens to P and Q when supply increases?

► Recall 39: What happens to P and Q when supply decreases?

► Recall 40: What happens to P and Q when demand decreases?

► Recall 41: Explain Tutorial Q1 in one sentence.

► Recall 42: Explain Tutorial Q2 in one sentence.

► Recall 43: Explain Tutorial Q3 in one sentence.

► Recall 44: What is short-run competitive equilibrium?

► Recall 45: What is long-run competitive equilibrium?

► Recall 46: Why does positive economic profit attract entry?

► Recall 47: Why does entry lower price?

► Recall 48: What is zero economic profit?

► Recall 49: Why does $P = AC$ in the long run?

► Recall 50: Why does $P = MC = AC$ imply minimum AC?

► Recall 51: What are the two tests for competitive equilibrium?

► Recall 52: Why might chocolate bars be approximated as competitive?

► Recall 53: What is the most common shift vs movement trap?

19. One-page cram sheet

Unit 8 Part B Cram Sheet

Core equations

$$Q_d = 40 - 2P$$
$$Q_s = 4P - 8$$

Equilibrium: $Q_d = Q_s \rightarrow P^* = 8, Q^* = 24$

Inverse D: $P = 20 - \frac{1}{2}Q$

Inverse S: $P = 2 + \frac{1}{4}Q$

Firm rules

Price-taking firm: $MR = P$

Profit max: $MR = MC \rightarrow P = MC$

Long run: $P = MR = MC = AC$

Shift rules

Event	P	Q
Demand ↑	↑	↑
Demand ↓	↓	↓
Supply ↑	↓	↑
Supply ↓	↑	↓

Must-write lines

- **Surplus:** $Q_s > Q_d \rightarrow$ sellers compete $\rightarrow$ price falls.
- **Shortage:** $Q_d > Q_s \rightarrow$ buyers compete $\rightarrow$ price rises.
- **Demand shift:** quantity demanded changes at every price.

- **Supply shift:** quantity supplied changes at every price.
- **Movement along curve:** caused by the good's own price changing.
- **Long run:** entry/exit eliminates economic profit under free entry/exit.

Deadly traps

- Do not say supply increased just because equilibrium quantity supplied rose.
- Do not say a price-taking firm lowers price to MC.
- Do not confuse WTP/WTB with actual market price.
- Do not confuse zero economic profit with no accounting profit.

21

20. Unit 8 Part B 80% readiness checklist

Before moving on to Unit 8 Part C or Unit 7 Part D, you should be able to complete these tasks without checking notes.

Readiness rule

You are ready when you can solve the algebra, draw the graphs, and explain the adjustment story out loud. Reading the answer and “recognising it” is not enough for 80%+ performance.